An Introduction to Intrinsic Fluid Force

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Abstract

In fluid dynamics, pressure is typically thought of as a force mediating the compression of a fluid. Throughout this paper we'll explore a geometric reformulation of pressure and its applications to various problems.

Intrinsic fluid force is the fictitious force enacted on fluid parcels moving through an intrinsically curved backdrop. The paths they take are the geodesics of the system; pressure and viscosity then manifest as curvature in the backdrop, instead of as forces themselves. This is similar in spirit to Arnold's 1966 paper on fluid flow as the geodesics of volume-preserving diffeomorphisms, now extended to the Navier-Stokes equations through a finite-dimensional metric.

In this paper, we establish two claims:

1. An existing flow's pressure and viscosity can always be represented by an intrinsic fluid force.
2. An existing flow's smoothness breaks if and only if the metric degenerates or contains discontinuities.

where an existing flow refers to any point at which $\mathbf{u} \neq 0$ and $\rho \neq 0$.

Intrinsic Fluid Force

Our first goal is to mathematically represent arbitrary pressures and viscosities as intrinsic fluid forces. We can begin by recalling the free geodesic equation:

$$\ddot{x}_i = -\Gamma_{ijk} u^j u^k \qquad \Gamma_{ijk} = \frac{1}{2}\left(\frac{\partial g_{ik}}{\partial x_j} + \frac{\partial g_{ij}}{\partial x_k} - \frac{\partial g_{jk}}{\partial x_i}\right)$$

where $\mathbf{u}$ is the velocity field. We can expand the LHS using the Navier-Stokes equation, which tells us that:

$$\ddot{x}_i = \frac{Du_i}{Dt} = -\frac{1}{\rho}\frac{\partial P}{\partial x_i} + \frac{1}{\rho}\frac{\partial}{\partial x_j}\left(\mu\frac{\partial u_i}{\partial x_j} + \mu\frac{\partial u_j}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(\left(\mu_b - \frac{2\mu}{3}\right)\frac{\partial u_k}{\partial x_k}\right) + F_i = f_i$$

for some $\mathbf{F}$, μ, and μ_b. Setting f_i equal to the Christoffel contraction:

$$f_i = -\Gamma_{ijk} u^j u^k \implies -f_i = \Gamma_{ijk} u^j u^k$$

Expanding the Christoffel:

$$\Gamma_{ijk}u^j u^k = \frac{1}{2}\left(\frac{\partial g_{ik}}{\partial x_j} + \frac{\partial g_{ij}}{\partial x_k} - \frac{\partial g_{jk}}{\partial x_i}\right)u^j u^k = \frac{1}{2}\left(\frac{\partial g_{ik}}{\partial x_j}u^j u^k + \frac{\partial g_{ij}}{\partial x_k}u^j u^k - \frac{\partial g_{jk}}{\partial x_i}u^j u^k\right)$$

Relabeling $j \leftrightarrow k$, the first two terms are equal:

$$\frac{\partial g_{ij}}{\partial x_k}u^j u^k - \frac{1}{2}\frac{\partial g_{jk}}{\partial x_i}u^j u^k = -f_i$$

This is the intrinsic fluid force equation, which is linear in derivatives of g_{ij} but depends on g^{ij} through u^i, making it quasi-linear in general. It relates pressure and viscosity to the metric, giving one equation per i, so the metric is underdetermined.

To ensure a solution exists, we'll treat this as a Cauchy problem. We'll have a smooth hypersurface with unit conormal ξ_i, and we'll prescribe g_{ij} along the surface as smooth initial data. Splitting the derivative of g_{ij} into a tangential part T_{ij} and unknown part $\xi_k h_{ij}$:

$$\frac{\partial g_{ij}}{\partial x_k} = T_{ijk} + \xi_k h_{ij}$$

for some h_{ij}. T_{ijk} is known, so when we substitute in we can split further into known and unknown pieces:

$$\frac{\partial g_{ij}}{\partial x_k}u^j u^k - \frac{1}{2}\frac{\partial g_{jk}}{\partial x_i}u^j u^k = \left(T_{ijk} + \xi_k h_{ij}\right)u^j u^k - \frac{1}{2}\left(T_{jki} + \xi_i h_{jk}\right)u^j u^k = -f_i$$

$$T_{ijk}u^j u^k + \xi_k h_{ij}u^j u^k - \frac{1}{2}T_{jki}u^j u^k - \frac{1}{2}\xi_i h_{jk}u^j u^k = -f_i$$

$$-f_i - T_{ijk}u^j u^k + \frac{1}{2}T_{jki}u^j u^k = \xi_k h_{ij}u^j u^k - \frac{1}{2}\xi_i h_{jk}u^j u^k$$

Contracting both sides with u^i:

$$-f_i u^i - T_{ijk}u^i u^j u^k + \frac{1}{2}T_{jki}u^j u^k u^i = -f_i u^i - \frac{1}{2}T_{ijk}u^i u^j u^k = \xi_k u^k h_{ij}u^i u^j - \frac{1}{2}\xi_i u^i h_{jk}u^j u^k$$

Denoting $\ell = h_{jk}u^j u^k$:

$$-f_i u^i - \frac{1}{2}T_{ijk}u^i u^j u^k = \xi_k u^k \ell - \frac{1}{2}\xi_i u^i \ell = \frac{1}{2}\xi_k u^k \ell$$

If $\xi_k u^k$ is nonzero, we can solve for the unknown ℓ:

$$\ell = h_{jk}u^j u^k = -\frac{2f_i u^i + T_{ijk}u^i u^j u^k}{\xi_k u^k}$$

so long as $\xi_k u^k$ is nonzero. ℓ is now purely in terms of known quantities, so we can move it to the LHS in our original equation:

$$-f_i - T_{ijk}u^j u^k + \frac{1}{2}T_{jki}u^j u^k + \frac{1}{2}\xi_i \ell = \xi_k u^k h_{ij} u^j$$

As long as $\xi_k u^k$ is nonzero, which is always locally arrangeable:

$$\frac{-f_i - T_{ijk}u^j u^k + \frac{1}{2}T_{jki}u^j u^k + \frac{1}{2}\xi_i \ell}{\xi_k u^k} = h_{ij}u^j = V_i$$

To prove existence, we need to show that some h_{ij} exists satisfying this equation. Trying:

$$h_{ij} = \frac{V_i u_j + u_i V_j}{u^k u_k} - \frac{u^l V_l u_i u_j}{(u^k u_k)^2}$$

Taking $h_{ij}u^j$:

$$h_{ij}u^j = \frac{V_i u_j u^j + u_i V_j u^j}{u^k u_k} - \frac{u^l V_l u_i u_j u^j}{(u^k u_k)^2} = V_i + \frac{u_i V_j u^j}{u^k u_k} - \frac{u^l V_l u_i}{u^k u_k} = V_i$$

This holds for any existing flow, which is where $u_i \neq 0$. With the transverse derivative found, we know a consistent $\partial g_{ij}/\partial x_k$ can exist, validating claim 1: a metric derivative always exists that can represent an existing flow.

Note that any solution to the intrinsic fluid force equation must be smooth unless a singularity occurs; a discontinuity in g_{ij} would spark a divergence in the metric side of the equation.

Suppose a discontinuity develops in P or $\mathbf{u}$. The Navier-Stokes side of the intrinsic fluid force equation will diverge, meaning the metric side must as well. Expanding u^i, we see one way how:

$$\frac{\partial g_{ij}}{\partial x_k}u^j u^k - \frac{1}{2}\frac{\partial g_{jk}}{\partial x_i}u^j u^k = \frac{\partial g_{ij}}{\partial x_k}g^{jp}g^{kq}u_p u_q - \frac{1}{2}\frac{\partial g_{jk}}{\partial x_i}g^{jp}g^{kq}u_p u_q$$

The intrinsic fluid force equation is quadratic in the inverse of g_{ij}. As a result, g_{ij} itself cannot diverge; instead, we need $\det g_{ij} \to 0$ so g^{ij} diverges. This corresponds to g_{ij} breaking down as a valid metric and losing it's positive-definiteness. This equation can also diverge through discontinuities in g_{ij}, so if g_{ij} is not smooth locally, the metric side will diverge. The classic Navier-Stokes problem can then be reformulated as: can the metric degenerate in finite time, or evolve discontinuously?

We can also prove the bidirectionality of the above statements. If the metric side diverges, the Navier-Stokes side must as well; this can happen through discontinuities in $\mathbf{u}$ and P, or $\rho = 0$. For an existing flow $\rho \neq 0$ and we can definitively validate claim 2: an existing flow breaks smoothness if and only if the metric degenerates or contains discontinuities.

Open Questions

What do phenomena like the energy cascade look like in intrinsic fluid forces? There are various contested or unexplained phenomena regarding fluid flow. Recasting these problems in terms of intrinsic fluid forces may shed light on their inner workings.

Can regularity be established from the intrinsic fluid force setup? Our construction relocates the regularity problem to proving the smoothness and validity of the metric based on the intrinsic fluid force equation. A regularity proof could be constructed via bootstrapping arguments or continuity.

What does the metric look like for standard environments? Whether the metric is natural for standard flows remains an open question. If the metrics are natural, intrinsic fluid force could stand as a physical interpretation of what's happening in fluids.

What happens at stagnation points? Intrinsic fluid force is an exclusively dynamic framework. The metric degenerates when $u_i = 0$ to maintain the intrinsic fluid force equation; finding out how the geometry acts at stagnation points would be a crucial step towards solving the regularity problem.